

Program : Diploma in Artificial Intelligence & Machine Language	
Course Code : 3347	Course Title: Python Programming Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 4 (L:1 T:0 P:3)	Periods per semester: 60

Course Objectives:

- Students should be able to use Python as a tool for solving computational problems

Course Prerequisites:

Topic	Course Title	Semester
Basic programming skills	Introduction to IT systems Lab	1
	Problem Solving & Programming	2
	Problem Solving & Programming Lab	2

Course Outcomes

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Develop simple programs using Python	8	Applying
CO2	Illustrate complex data types and functions in Python	15	Applying
CO3	Illustrate exception handling & object oriented programming concepts in Python	17	Applying
CO4	Use the modules Numpy, Matplotlib, and Pandas to process data	17	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3- Strongly mapped, 2-Moderately mapped, 1- Weekly Mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop simple programs using Python		
M1.01	Develop programs using variables and data types.	2L+6P	Applying
M1.02	Develop programs using selection control structures		Applying
M1.03	Make use of iterative control structures		Applying
CO2	Illustrate complex data types and functions in python		
M2.01	Illustrate string operations in Python	1L+6P	Applying
M2.02	Develop programs using Lists, Tuples, Sets, Dictionaries		Applying
M2.03	Demonstrate file operations	2L+6P	Applying
M2.04	Develop programs using functions.		Applying
	Lab Exam – I	1.5	
CO3	Illustrate exception handling & object oriented programming concepts in Python		
M3.01	Demonstrate Exception handling for real-time applications. (divide by zero error, voter’s age validity, student mark range validation)	1L+3P	Applying
M3.02	Develop programs using class and objects		Applying

M3.03	Demonstrate the concept of constructor	2L+6P	Applying
M3.04	Demonstrate the concept of inheritance		
CO4	Use the modules Numpy, Matplotlib, and Pandas to process data		
M3.01	Use NumPy to create arrays	1L+3P	Applying
M3.02	Implement mathematical operations using NumPy arrays		Applying
M3.03	Demonstrate matrix operations using NumPy		Applying
M3.04	Develop a python code to read a .csv file using panda module and print the first and last five lines of a file.	1L+6P	Applying
M3.05	Use Matplotlib to plot lines		Applying
M3.06	Use Matplotlib to draw geometric shapes		Applying
M3.07	Open ended experiments	6	Applying
	Lab Exam – II	1.5	

**** Suggested Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Compute the grades of students in a class and visualize the result

Text / Reference:

T/R	Book Title / Author
R1	Swapnil Saurav, <i>Learn and Practice Python Programming</i> , Eka Publishers, 2020
R2	Introduction to Computer Science Using Python - A Computational Problem - Solving Focus - Chales Dierbach Wiley India Pvt. Ltd
R3	Yashavant Kanetkar, Aditya Kanetkar, “Let us Python, BPB publication, 1st Edition, 2019
R4	Allen B. Downey, “Think Python : How to Think like a Computer Scientist”, 2nd Edition, O’Reilly Publishers, 2016.
R5	R1 Paul Deitel and Harvey Deitel, “Python for Programmers”, Pearson Education, 1st Edition, 2021.

Online Resources:

Sl. No.	Website Link
1	https://docs.python.org/3/
2	https://www.programiz.com/python-programming